

Modeling Bitcoin Returns and Volatility: The Role of Mining Difficulty

Financial Econometrics Final Exam Project

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1. Introduction

Bitcoin introduces a unique asset class with a fixed supply schedule governed by an algorithm that adjusts mining difficulty every 2016 blocks—approximately every two weeks. This mechanism is designed to stabilize issuance, but may also generate short-term feedback effects with price, particularly when demand fluctuates. Unlike traditional markets where supply reacts flexibly, Bitcoin's protocol responds with lagged and rule-based difficulty adjustments that could amplify price movements.

This paper investigates whether mining difficulty exerts a causal effect on Bitcoin's USD exchange rate by acting as a short-term supply friction. As illustrated in Figures 1 and 2, a demand shock shifts the equilibrium price and quantity. The subsequent difficulty adjustment—by reducing or increasing mining difficulty—can shift the effective supply curve. The original price movement gets reinforced from a classic price reaction on demand shift, moving from the original equilibrium Q_1 and P_1 to Q_2^* and P_2^* in both Figure 1 and 2, to Q_2 and P_2 , due to Bitcoin's lagged mining difficulty adjustment that acts as supply shift. One should note that the mechanism theoretical succeeds in keeping Q_2 equal to Q_1 , however prices are much more amplified than under normal market adjustment: In Figure 1 instead of dropping to P_2^* they drop to even lower P_2 as a result of demand decrease and in Figure 2 they not only increase to P_2^* but to P_2 as a result of demand increase.

The null hypothesis tested is that difficulty does not Granger-cause Bitcoin returns, implying a purely passive role in the pricing process. If the null is not rejected, this may indicate that mining difficulty plays no causal role—or that the protocol is functioning as intended: stabilizing quantity and letting prices absorb demand shocks, but with no immediate reflection in the USD exchange rate if miners delay converting rewards into fiat.

The empirical analysis proceeds in four steps: First, an ARIMA model establishes a benchmark for Bitcoin returns. Second, an ARIMAX model incorporates macroeconomic, behavioral, and blockchain variables to test their predictive power. Third, Granger causality is tested between mining difficulty and returns. Finally, a GARCH(1,1) model evaluates volatility clustering and the explanatory power of conditional variance.

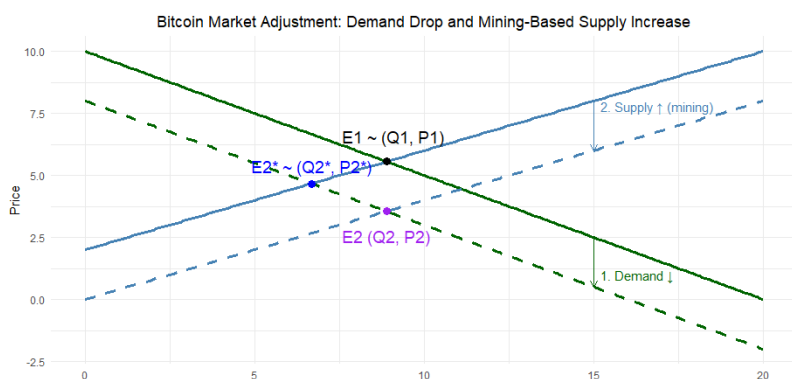


Figure 1: Visualization of theoretical market reaction on Bitcoin specific supply increase from demand decrease

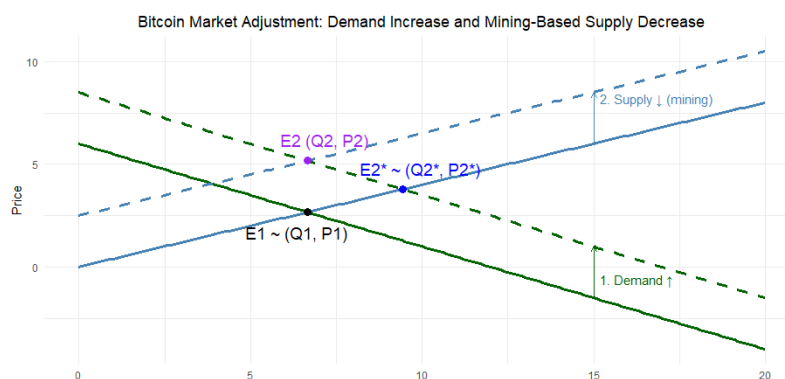


Figure 2: Visualization of theoretical market reaction on Bitcoin specific supply decrease from demand increase.

2. Literature Review

This paper draws on two main strands of empirical research on Bitcoin: volatility modeling via GARCH processes and return predictability through macroeconomic, behavioral, and blockchain-native variables.

Katsiampa (2017) and Dyhrberg (2016) show that GARCH(1,1) models effectively capture the volatility clustering observed in Bitcoin returns, justifying their use for modeling time-varying risk. Dyhrberg, in particular, treats Bitcoin as a hybrid asset—part commodity, part currency—highlighting its asymmetric volatility responses to market events.

On the return side, studies like Corbet et al. (2018) argue that Bitcoin clearly is in a bubble phase while Bouri et al. (2020) use a wavelet-based framework to show that Bitcoin exhibits weak co-movement with stock markets across time and frequency scales, reinforcing its role as a portfolio diversifier. Their analysis also incorporates value-at-risk (VaR) metrics, revealing that Bitcoin may offer superior tail-risk hedging compared to gold and commodities.

However, relatively little attention has been paid to Bitcoin's internal supply adjustment mechanism—mining difficulty. Guizani and Nafti (2019) find that difficulty significantly predicts Bitcoin prices in the short term, as it reflects production costs. Yet they also note that this effect diminishes over time due to technological advances. This project builds on their findings by empirically testing whether mining difficulty amplifies price movements through lagged supply adjustments, as theoretically depicted in Figures 1 and 2.

By incorporating mining difficulty into ARIMAX and Granger causality models—alongside traditional macroeconomic and sentiment variables—this paper expands on the existing literature and contributes a novel perspective on the endogenous mechanics of Bitcoin pricing.

3. Data

The dataset used in this project consists of daily time series spanning from January 1, 2017 to December 31, 2024. The primary variable of interest is the daily return of Bitcoin, calculated from the BTC-USD adjusted closing price obtained via Yahoo Finance. Volatility is modeled using these return series.

To evaluate the influence of blockchain-native dynamics, the dataset incorporates daily mining difficulty, hash rate, and transaction volume data extracted from Blockchain.com's API. Additionally, external financial variables including Ethereum returns (ETH-USD), S&P 500 index returns, and the U.S. 10-year Treasury yield (DGS10) from the Federal Reserve Economic Data (FRED) are included to capture macroeconomic influence.

Behavioral data is added via Google Trends, using the search term "Bitcoin". The trends data,

initially recorded monthly, was forward-filled to a daily frequency to align with the rest of the dataset. After merging all series, the final dataset was cleaned to remove initial NA values and limited to a uniform period for modeling purposes.

4. Methodology

This study employs a structured econometric approach using three key time series models: an ARIMA framework for baseline analysis, an ARIMAX model with exogenous regressors, and Granger causality analysis to explore temporal dependencies between BTC-USD returns and blockchain fundamentals. Finally, a GARCH(1,1) model is employed to capture volatility dynamics.

4.1 Benchmark ARIMA Modelling and Random Walk Hypothesis

The analysis begins with an ARIMA model to assess the time series structure of BTC-USD returns and to test the weak-form efficient market hypothesis (EMH). Specifically, the random walk hypothesis implies that the best predictor of tomorrow's price is today's price, i.e., returns are serially uncorrelated and follow a white noise process:

$$R_{BTC_t} = \epsilon_t, \quad \epsilon_t \sim \text{i.i.d.}$$

This corresponds to an ARIMA(0,0,0) with zero mean. Alternatively, if prices follow a random walk, this paper expects the first-differenced price series to follow an ARIMA(0,1,0) process:

$$P_t = P_{t-1} + \epsilon_t$$

Lag selection is determined using the Bayesian Information Criterion (BIC), and stationarity is evaluated via the Augmented Dickey-Fuller (ADF) test.

4.2 ARIMAX and Difficulty Hypothesis

To investigate whether blockchain-related and macroeconomic variables can explain variations in BTC-USD returns, an ARIMAX model is estimated. This model extends the ARIMA structure to include exogenous regressors x_{t-k} :

$$y_t = \mu + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \sum_{k=1}^K \delta_k x_{t-k} + \epsilon_t$$

The regressors include mining difficulty, transaction volume, Ethereum returns, S&P 500 returns, the 10-year U.S. Treasury yield, and Google Trends search interest. The ARIMAX model follows the logic and form of the Autoregressive Distributed Lag (ADL) model.

4.3 Granger Causality Analysis

A Granger causality framework is used to explore whether changes in mining difficulty help predict BTC-USD returns. While a full VAR model is often employed, estimation challenges such as multicollinearity (e.g., between difficulty and hash rate) required a simplified approach. Instead, direct Granger tests are performed using the following form:

$$y_t = \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \delta_j x_{t-j} + \epsilon_t$$

The null hypothesis of no Granger causality is tested using nested F-tests, with lag order selected via AIC and BIC. Series found to be non-stationary are first-differenced before inclusion.

4.4 GARCH(1,1) Model for Volatility Dynamics

To model volatility clustering — a common feature in financial time series — a GARCH(1,1) model is used. In this framework, return innovations are assumed to follow:

$$u_t = \sigma_t \cdot \epsilon_t, \quad \epsilon_t \sim N(0,1)$$

with the conditional variance evolving as:

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Where σ_t^2 denotes the conditional variance, ϵ_{t-1}^2 the lagged squared residual from the mean equation, and σ_{t-1}^2 the lagged conditional variance.

This specification captures the tendency for periods of high (or low) volatility to persist over time. GARCH(1,1) is chosen due to its empirical success, parsimony, and interpretability. Model residuals and squared residuals are analyzed through their ACF and PACF plots to justify the presence of ARCH effects prior to estimation.

To evaluate the adequacy and robustness of GARCH models, three main diagnostic tests are employed. First the Ljung-Box Q-test: This test examines whether residuals exhibit serial correlation, indicating if they behave like white noise. It is based on the statistic:

$$Q = T(T+2) \sum_{i=1}^m \frac{p_i^2}{T-i}$$

where p_i represents autocorrelation at lag i , and T is the sample size. A non-significant p-value suggests no residual autocorrelation, supporting a properly specified mean model.

Second, an ARCH LM test: This test specifically checks for remaining autoregressive conditional heteroskedasticity (ARCH) effects in residuals. It uses the auxiliary regression:

$$u_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \dots + \alpha_p u_{t-p}^2 + \epsilon_t$$

A non-significant joint test of coefficients $\alpha_1, \dots, \alpha_p$ indicates no remaining ARCH effects, confirming a correctly specified variance equation.

Third, a Nyblom Stability test: This assesses parameter stability over the sample period, ensuring that the estimated model parameters do not vary significantly across time. A stable parameter set strengthens the reliability of the GARCH estimates.

5. Results

This section presents the results of three key models applied to the BTC-USD return series: a benchmark ARIMA model, an extended ARIMAX model including macroeconomic and blockchain-related variables, an exploratory Granger causality test assessing dynamic interdependencies, and a GARCH(1,1) volatility model.

5.1 ARIMA

The benchmark ARIMA model selected by the Bayesian Information Criterion (BIC) on the returns of BTC-USD was ARIMA(0,0,0) with zero mean. This implies that the return series is best modelled by a multiplicative random walk, hence supporting weak form efficient market hypothesis:

$$R_{BTC} = \frac{P_{BTC_t} - P_{BTC_{t-1}}}{P_{BTC_{t-1}}} = 0 + \epsilon_t$$

$$R_{BTC} * P_{BTC_{t-1}} = P_{BTC_t} - P_{BTC_{t-1}}$$

$$(0 + \epsilon_t) * P_{BTC_{t-1}} + P_{BTC_{t-1}} = P_{BTC_t}$$

$$P_{BTC_t} = P_{BTC_{t-1}} * (1 + \epsilon_t)$$

→ coefficient = 1 with ϵ_t being white noise (see Figure 3)

→ multiplicative random walk

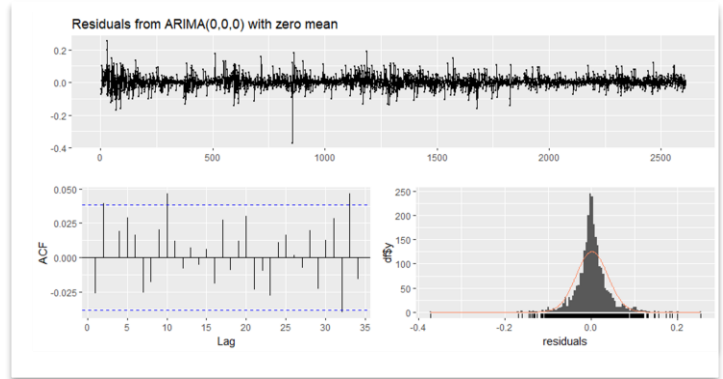


Figure 3: Residuals of ARIMA(0,0,0) on BTC-USD return time series

As a quick recap, that suggests that all information of the past (weak market efficiency) is priced in and new information results in a random change in price which make them impossible to forecast.

To see whether the textbook random walk model, the additive random walk, also finds support in the data the `auto.arima` function in R has been run on BTC-USD price, which suggested an ARIMA(0,1,0) model, so a first differencing, confirming the random walk model:

$$\Delta P_{BTC_t} = P_{BTC_t} - P_{BTC_{t-1}} = \epsilon_t$$

$$P_{BTC_t} = P_{BTC_{t-1}} + \epsilon_t$$

To assess the out-of-sample forecasting performance of the benchmark ARIMA(0,0,0) model, the dataset was divided into a training set (2017–2023) and a test set (January to December 2024). The model was estimated on the training data, and one-step-ahead forecasts were recursively generated for the test period. As shown in Figure 4, the forecasted returns (blue line) remain constant at zero due to the model's structure, while the actual returns (red line) display greater variability around the forecast. The visualization confirms that the ARIMA(0,0,0) model with zero mean simply projects future returns as white noise around zero, reflecting the underlying random walk assumption.

Despite its simplicity, the model performs reasonably well in forecasting accuracy, yielding a root mean squared error (RMSE) of 0.0281 and a mean absolute error (MAE) of 0.0204 on the test set. The constant forecast and wide prediction intervals reflect the model's neutrality and high uncertainty in predicting BTC-USD returns. This aligns with the efficient market hypothesis, suggesting that return innovations are unpredictable given available information.

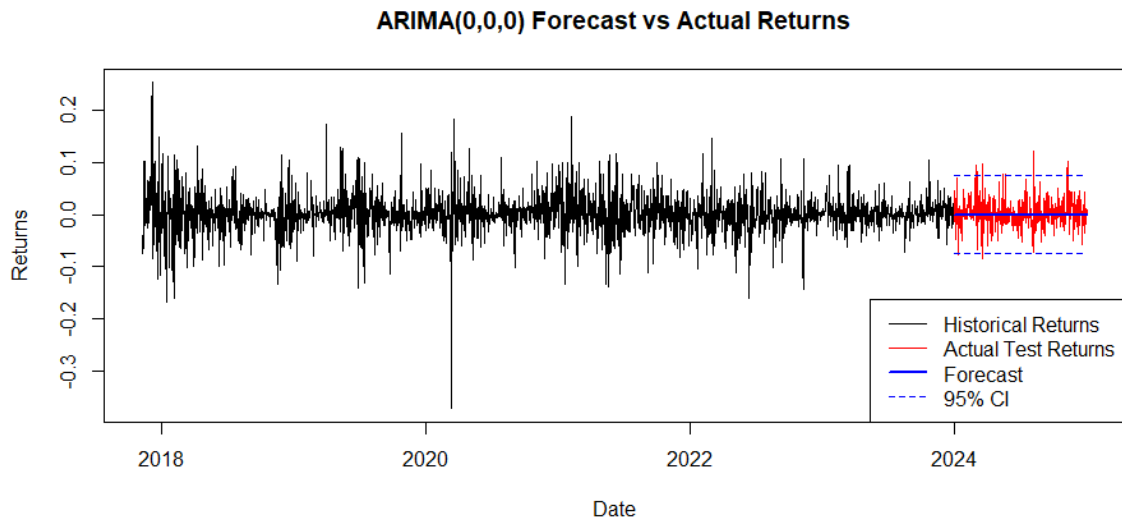


Figure 4: ARIMA(0,0,0) Model of BTC-USD Return Series vs. Actual BTC-USD Return Series

5.2 ARIMAX

The ARIMAX model incorporated additional explanatory variables including difficulty, transaction volume, Ethereum returns, S&P 500 returns, 10-year U.S. Treasury yield, and Google search trends. Due to the extremely high correlation (0.99) between hash rate and difficulty, hash rate was excluded from the model to avoid multicollinearity. Additionally, difficulty is a lagged, endogenous function of hash rate and market demand. It is recalculated by the Bitcoin protocol every 2016 blocks to maintain a steady 10-minute block interval. Its formula can be expressed conceptually as:

$$Difficulty_t = Difficulty_{t-1} \times \frac{Expected\ Time}{Actual\ Time}$$

This adjustment mechanism is briefly mentioned in the original Bitcoin white paper by Satoshi Nakamoto, where it is stated that the network adjusts the difficulty to maintain a constant block interval, targeting an average of one block every ten minutes. The white paper outlines that if blocks are generated too quickly or slowly, the difficulty is recalibrated accordingly to restore the desired output (p. 3, Nakamoto, 2008).

Because difficulty adjusts with a lag, its use as an explanatory variable is less prone to simultaneity bias than hash rate, which adjusts more contemporaneously with returns. Therefore, excluding hash rate not only reduced multicollinearity but also mitigated potential endogeneity concerns.

$$R_{BTC_t} = \begin{matrix} 0.0016 \\ (0.0005) \end{matrix} + \begin{matrix} 0.0004 \cdot Dif_t \\ (0.0007) \end{matrix} - \begin{matrix} 0.0003 \cdot V_{BTC_t} \\ (0.0005) \end{matrix} + \begin{matrix} 0.0281 \cdot R_{ETH_t} \\ (0.0005) \end{matrix} + \begin{matrix} 0.0017 \cdot R_{S\&P} \\ (0.0005) \end{matrix} - \begin{matrix} 0.0001 \cdot TY_{10y} \\ (0.0007) \end{matrix} - \begin{matrix} 0.0007 \cdot GS_{BT} \\ (0.0005) \end{matrix} + \begin{matrix} 0.1087 \varepsilon_{t-1} \\ (0.0196) \end{matrix} + \varepsilon_t$$

$t =$ at time t ; $t - 1 =$ one day before time t ; R_{BTC} = Return of Bitcoin – USD exchange rate;

Dif = Difficulty; V_{BTC} = Bitcoin Volume; R_{ETH} = Return Ethereum;

TY_{10y} = 10 years treasury yield; GS_{BT} = Google Searches for Bitcoin; ε = error term

Standard deviations in brackets

Among the explanatory variables, Ethereum returns were highly statistically significant and positive, indicating a strong co-movement or spillover effect between the broader crypto market and Bitcoin. The coefficient of S&P 500 returns was also positive and statistically significant, suggesting a modest connection to traditional markets.

In contrast, several variables — including mining difficulty, transaction volume, 10-year Treasury yield, and Google Trends searches — were statistically insignificant. These results suggest that, in this specification, they offer no additional predictive power for daily BTC-USD returns. Specifically, the original hypothesis that mining difficulty would be a key driver of BTC-USD price dynamics is not supported by the data, as its coefficient is statistically insignificant.

To confirm this, the same ARIMAX model was estimated using the BTC-USD price level (instead of returns) as the dependent variable. This resulted in similar patterns of significance and insignificance, and the underlying ARIMA structure in that case was an ARIMA(1,1,0), aligning with a random walk with drift interpretation.

It is important to note that all variables were standardized prior to estimation using `scale()`, because the model would not converge otherwise. As a result, the numerical magnitudes of the coefficients are not directly interpretable in economic terms. However, the signs and statistical significance of the coefficients still offer valuable insight into the relationships between BTC-USD and these explanatory variables.

5.3 Granger Causality

To assess whether changes in mining difficulty Granger-cause BTC-USD returns, a Granger causality test was conducted. The Granger causality framework tests whether past values of one variable (in this case, mining difficulty) contain information that helps predict another variable (BTC-USD returns) beyond what is contained in its own past values. The null hypothesis (H_0) states that lagged values of mining difficulty do not Granger-cause BTC-USD returns, meaning that difficulty carries no predictive power for returns.

Before performing the Granger test, the stationarity of the involved time series was evaluated using the Augmented Dickey-Fuller (ADF) test. BTC-USD returns and Google search volume rejected the null hypothesis of a unit root, indicating stationarity at conventional significance levels. However, both mining difficulty and hash rate exhibited non-stationarity and were therefore differenced to achieve stationarity.

The model specifications compared a restricted model (Bitcoin-USD returns regressed only on their own lags) against an unrestricted model (Bitcoin-USD returns regressed on their own lags plus lagged values of differenced mining difficulty series).

To account for the protocol mechanism of Bitcoin's difficulty adjustment — which occurs every 2016 blocks, approximately every two weeks — a Granger causality test was performed using 20 lags. This longer lag structure captures potential delayed effects if the adjustment process takes longer due to slower block production. The F-statistic for the joint significance of lagged difficulty terms in this case was 0.8093 with a p-value of 0.7047. While the p-value slightly improved compared to a four-lag model with F-statistic of 0.2595 and corresponding p-value of 0.9039, it remains substantially above conventional significance thresholds. Consequently, there is still no statistical evidence that mining difficulty Granger-causes Bitcoin-USD returns and this paper fails to reject the Null hypothesis. Even more lags were tested to see whether a delayed exchange of Bitcoin to Dollar might only then show its effect as hypothesised in Figure 1 and 2, but also these tests remained insignificant.

To establish Granger Causality, one also has to test whether the BTC-USD returns cause the differenced difficulty series. They do not. So, Granger Causality is generally not found.

5.4 GARCH

Following the estimation of an ARIMA(0,0,0) model, residual diagnostics revealed pronounced signs of volatility clustering in the BTC-USD return series, as visible in the conditional heteroskedasticity pattern observed in the residual plot (Figure 3).

The variance of the residuals appears to fluctuate over time, with periods of elevated and subdued volatility occurring in clusters—a hallmark characteristic of financial time series. While the ACF of residuals shows little autocorrelation at lower lags, both the residual time series and the histogram indicate non-constant variance. This suggests that the series may benefit from a volatility model such as GARCH. Further evidence is provided in Figure 5, where the partial autocorrelation function (PACF) of the squared residuals exhibits several significant spikes. These persistent autocorrelations in squared residuals strongly indicate the presence of autoregressive conditional heteroskedasticity, thus validating the application of a GARCH(1,1) model. The estimation results from the equation in section 4.4 yielded:

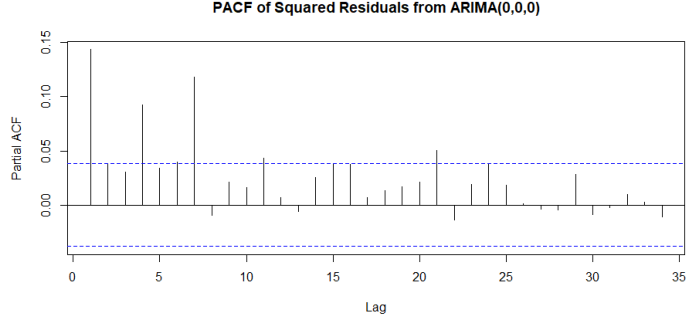


Figure 5: PACF of Squared Residuals from ARIMA(0,0,0)

$$\sigma^2_t = 0.000059 + 0.0817 \cdot \varepsilon_{t-1}^2 + 0.8755 \cdot \sigma_{t-1}^2$$

Standard deviations in brackets (0.000024) (0.030592) (0.038644)

All parameters were statistically significant at the 1% level, confirming the presence of strong volatility clustering and persistence. The high sum of $\alpha_1 + \beta_1 = 0.9572$ indicates that shocks to volatility decay slowly, a typical feature in financial return series.

Diagnostic tests confirm the adequacy of the estimated GARCH(1,1) model. The Ljung-Box tests applied to standardized and squared standardized residuals indicate no significant serial correlation (p-values well above the 5% significance threshold). Thus, the residuals behave as white noise, validating the chosen mean specification.

The ARCH LM tests reinforce this conclusion, showing no remaining autoregressive conditional heteroskedasticity in squared residuals, confirming that the volatility dynamics are fully captured by the GARCH(1,1) model. Lastly, the Nyblom stability test statistics remain below critical thresholds, indicating robust parameter estimates that do not significantly vary over the sample period.

Together, these results demonstrate that the fitted GARCH(1,1) model accurately captures the volatility clustering and conditional variance dynamics of BTC-USD returns.

To assess the out-of-sample volatility forecasting performance of the GARCH(1,1) model, the dataset was divided into a training set (up to 2023-12-31) and a test set (from 2024-01-01 onward). The GARCH(1,1) model was estimated on the training data, consistent with earlier results showing strong volatility clustering and significant ARCH and GARCH parameters. The

conditional standard deviation forecast $\hat{\sigma}_t$ was extracted and squared to obtain $\hat{\sigma}_t^2$, which serves as the model's forecasted volatility. Because the ARIMA(0,0,0) model assumes zero mean, realized volatility was proxied by squaring the actual returns in the test set: $R_t^2 \approx (R_t - E[R_t])$.

Figure 6 displays the resulting plot comparing forecasted conditional variance (blue line) with realized volatility (red line). While the GARCH model forecasts relatively stable and slowly adjusting volatility, realized volatility exhibits more erratic and spiky behavior—highlighting the challenge of capturing extreme volatility bursts with a standard GARCH(1,1) specification. Nonetheless, the forecasted volatility tracks the underlying baseline level reasonably well. The forecasting performance, evaluated via Mean Squared Error (MSE), yielded a value of 2.97×10^{-6} , indicating a modest deviation between predicted and actual volatility. When using the precise realised volatility and calculating the exact mean, the MSE is approximately 2.89×10^{-6} .

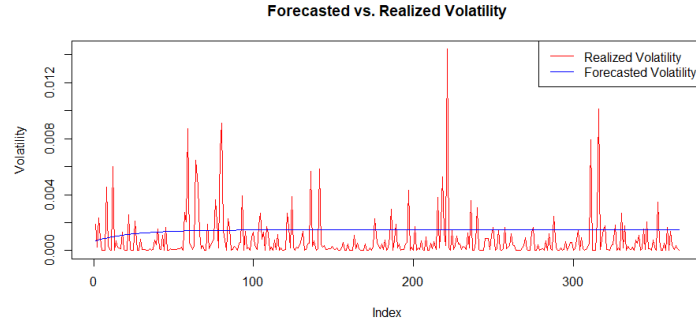


Figure 6: Realized volatility with exact mean plotted against forecasted volatility of the Garch(1,1) model.

6. Conclusion

This paper investigated whether Bitcoin's mining difficulty adjustment mechanism has a causal or amplifying effect on its USD exchange rate returns and volatility. Using a four-part time series approach—ARIMA, ARIMAX, Granger causality, and GARCH modelling—the study aimed to assess both return predictability and volatility behavior from 2017 to 2024.

The ARIMA(0,0,0) model supported the random walk hypothesis, indicating that BTC-USD returns are serially uncorrelated and consistent with weak-form market efficiency. When extending the model with macroeconomic, behavioral, and blockchain-native regressors, the ARIMAX results showed no significant predictive power for mining difficulty, though variables like Ethereum and S&P 500 returns were partially informative.

Granger causality tests confirmed these findings, showing no evidence that changes in mining difficulty cause changes in BTC-USD returns, even when extended lags were applied to reflect protocol adjustment cycles. This suggests that while difficulty regulates supply internally, it does not translate into short-term pricing effects at the exchange rate level.

Finally, GARCH modelling revealed strong volatility clustering in returns, reinforcing the importance of modelling conditional variance even when mean predictability is limited. Overall, the results suggest that the protocol may be working as intended—stabilizing supply without directly distorting return behavior—though further research could explore nonlinear effects or alternative data frequencies.

7. References

- Bouri, E., Shahzad, S., Roubaud, D., Kristoufek L., & Lucey, B. (2020). Bitcoin, gold and commodities as safe havens for stocks: New insight through wavelet analysis. *The Quarterly Review of Economics and Finance*, 77, 156–164. <https://doi.org/10.1016/j.qref.2020.03.004>
- Corbet, S., Lucey, B., & Yarovaya, L. (2018). Datestamping the Bitcoin and Ethereum bubbles. *Finance Research Letters*, 26, 81–88. <https://doi.org/10.1016/j.frl.2017.12.006>
- Dyhrberg, A. H. (2016). Bitcoin, gold and the dollar – A GARCH volatility analysis. *Finance Research Letters*, 16, 85–92. <https://doi.org/10.1016/j.frl.2015.10.008>
- Guizani, S., & Nafti, I. K. (2019). The determinants of Bitcoin price volatility: An investigation with ARDL model. *Procedia Computer Science*, 164, 233–238. <https://doi.org/10.1016/j.procs.2019.12.177>
- Katsiampa, P. (2017). Volatility estimation for Bitcoin: A comparison of GARCH models. *Economics Letters*, 158, 3–6. <https://doi.org/10.1016/j.econlet.2017.06.023>
- Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*. <https://bitcoin.org/bitcoin.pdf>
- Bitcoin Wiki. (n.d.). *Difficulty*. Retrieved April 29, 2025, from <https://en.bitcoin.it/wiki/Difficulty>